Architecture Design Document

Project 6: Student Performance Analysis System

1. Problem Statement

Educational institutions often face challenges in identifying the underlying causes of student underperformance. Raw academic grades (G1, G2, G3) are rarely integrated with external socio-economic or behavioral factors, making it difficult for educators to intervene effectively before a student fails.

2. System Objectives

- **Database Structure:** To develop a structured PostgreSQL database that handles both academic and demographic data.
- **Analytical Capability:** To provide tools for analysts to filter and correlate performance with variables like study time and parental education.
- **Actionable Insights:** To automate the generation of insights that assist in making data-driven educational decisions.

3. Architectural Pattern: MVC

The system is designed using the MVC architectural pattern to ensure a clean separation of concerns:

- **Model:** Handles the data logic and the PostgreSQL database schema consisting of 7 normalized tables.
- **View:** The React-based user interface that displays the dashboard, analytical charts, and report filters.
- **Controller:** The Node.js backend that processes requests, such as CSV file uploads and SQL query execution.

4. Use Case Analysis & Interactions

- **School Admin:** Interacts with the system to import and validate raw CSV data and manage database records.
- **Educational Analyst:** Interacts with the system to apply filters to socio-economic factors and generate performance insights.

5. Technology Stack

Component	Technology

Database	PostgreSQL
Backend	Node.js & Express
Frontend	React.js